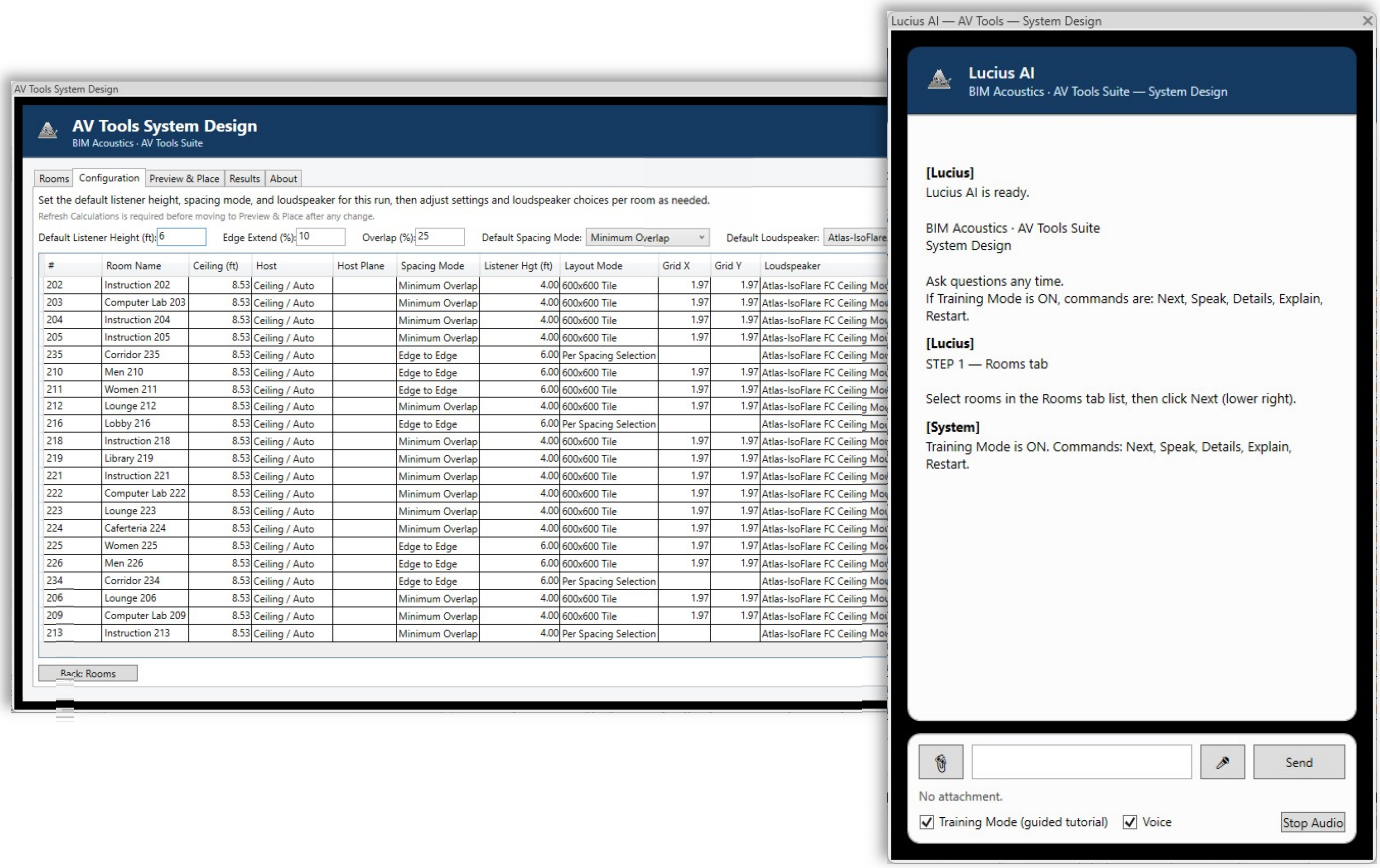
BIM Acoustics, LLC | 411 Forest Acres Cir.
Walhalla, SC 29691

504-401-4983 | www.bimacoustics.net

BIM Acoustics AVTools System Designer



Distributed Ceiling Loudspeaker Layout for Autodesk Revit

Version 1.2.1

J. Stevens BIM Acoustics

Overview

AVTools System Designer is a professional Revit add-in designed to automate distributed ceiling loudspeaker layout directly within Building Information Modeling (BIM) workflows.

The tool enables AV consultants, system designers, and integrators to generate consistent loudspeaker layouts across multiple rooms while maintaining clear engineering intent and coordination with architectural models.

By working directly within Autodesk Revit, System Designer eliminates much of the manual drafting, layout calculation, and coordination traditionally required when designing distributed audio systems in large facilities.

Key Capabilities

Automated Loudspeaker Layout

System Designer generates distributed ceiling loudspeaker layouts using established spacing models and configurable design parameters.

Layouts can be applied consistently across multiple rooms, avoiding manual copy-paste workflows and ensuring repeatable design logic.

Revit-Native Workflow

The add-in operates entirely inside Autodesk Revit, using model geometry such as:

- Room boundaries
- Ceiling conditions
- Host geometry

This allows loudspeaker layouts to remain coordinated with architectural and MEP models.

Tap Setting Guidance

Designers can define a target sound pressure level and receive recommended loudspeaker tap settings based on the selected design parameters.

All tap settings remain **designer-controlled**, ensuring engineering decisions remain transparent.

Coverage Visualization

System Designer provides **first-order ISO-line visualization** that allows designers to evaluate relative coverage uniformity and spacing effectiveness within the Revit model.

This early-stage visualization supports rapid layout refinement before detailed acoustic modeling.

Multi-Room Layout Consistency

Large facilities often contain dozens or hundreds of rooms requiring distributed audio systems.

System Designer allows layouts to be generated consistently across multiple rooms while preserving spacing logic and placement rules.

Lucius AI Assistant

System Designer includes **Lucius AI**, an embedded assistant that provides contextual workflow guidance within the Revit environment.

Lucius can assist users with topics such as:

- Spacing models
- Loudspeaker overlap strategy
- Tap setting considerations
- Coverage mapping concepts
- System Designer workflow guidance

The assistant is designed to help designers understand the engineering intent behind layout decisions while working inside the model.

What's New in Version 1.2.1

Version **1.2.1** introduces several improvements focused on stability and compatibility.

Linked Model Stability Improvements
Enhanced reliability when working with linked Revit models and tile-centered ceiling grids.

Project-Unit-Aware Reporting
System Designer now reflects the active Revit project length units in column headers and numeric reporting, supporting both **Imperial and Metric workflows**.

Revit 2026 Compatibility
Updates to shared parameter binding and category handling ensure compatibility with Autodesk Revit 2026.

JSBA Control Data Schedule Improvements
Enhanced multi-category schedule creation and population for project data reporting.

Platform Support

Autodesk Revit Versions

- Revit 2022
- Revit 2023
- Revit 2024
- Revit 2025
- Revit 2026

Availability

AVTools System Designer is available through **J. Stevens BIM Acoustics**.

Subscription licensing options are available.

More information:

https://www.bimacoustics.net

About J. Stevens BIM Acoustics

J. Stevens BIM Acoustics is a consulting and software development firm focused on advancing acoustics and audiovisual system design through Building Information Modeling (BIM). The company develops tools that integrate AV system design directly into Revit-based project environments, enabling model-centric workflows for audio and acoustics design.

The firm was founded by **AV consultant Jerrold Stevens**, who has spent decades working in acoustics, sound reinforcement, and audiovisual system design. Stevens has also taught professional courses on sound system design and loudspeaker arrays at major industry events including InfoComm.

Stevens currently serves on **AVIXA's BIM Standards Subcommittee**, contributing to the development of industry standards for integrating audiovisual systems within BIM-based design workflows.

Media Contact

Jerrold W. Stevens
J. Stevens BIM Acoustics
https://www.bimacoustics.net
j.stevens@bimacoustics.net